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SIMATS ENGINEERING
CO2 - ASSESSMENT TOOL 3

Class Test

COURSE NAME: Deep Learning
Faculty : Dr. Arul Raj A.M

COURSE CODE: MLA04
Slot :

COURSE OUTCOME COVERED

CO 2: Analyze Machine Learning and Deep Learning models by evaluating neural network architectures, representation learning, activation functions, unsupervised learning techniques, and their real-world applications. (L4)

CO Weightage : 15%

Assessment Tool Weightage: 20%

Duration: 90 minutes

Marks: 25

Code of Conduct:

I N. Jeevan Sai (192525169) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

N. Jeevan Sai

Signature of the student:

Name of the student:

Criteria	Conceptual Understanding (2.5)	Linkages (2.5)	Organization (2)	Integration of Literature (2)	Clarity & Presentation (1)	Total (10)
Weightage	25 %	25%	20%	20 %	10 %	100%
Q1						
Q2						
Q3						
Q4						
Q5						
Total						

Class Test

CO2-AT3

QUESTIONS: 05

Total Marks:25

1. Differentiate between Machine Learning and Deep Learning with respect to architecture, data requirements, feature extraction and applications. Give two real-world applications of each.
2. Explain the concept of **Representation Learning**. Discuss how the **width** and **depth** of a neural network influence its learning capability, computational complexity and model performance.
3. Compare **ReLU**, **Leaky ReLU (LReLU)** and **ELU (Exponential Linear Unit)** activation functions. Explain their mathematical behavior, advantages, disadvantages and suitable use cases.
4. Explain the concept of **unsupervised learning** in neural networks. Describe the working principle of **Restricted Boltzmann Machines (RBMs)** and **Autoencoders** and compare their objectives and applications.
5. Explain the concept of Autoencoder and its uses in Deep learning .

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Concept Identification	25%	Identifies all key concepts from literature accurately and comprehensively	Identifies most key concepts with minor omissions	Identifies limited concepts; some are irrelevant or missing	Fails to identify essential concepts
Linkages	25%	Establishes clear, meaningful, and accurate relationships between concepts	Shows correct relationships with minor gaps	Relationships are weak, unclear, or partially incorrect	No meaningful relationships established
Organization	20%	Concepts are well-structured hierarchically with logical flow and grouping	Mostly organized with minor structural issues	Some organization but lacks clear hierarchy	No logical organization or structure
Integration of Literature	20%	Integrates multiple studies effectively to show connections and comparisons	Shows some integration of literature	Limited integration; mostly isolated concepts	No integration of literature evidence
Clarity	10%	Concept map is clear, neat, visually structured, and	Generally clear with minor	Clarity is reduced; difficult to	Poorly presented and hard to

Jeevan Sai
192525169.

Machine learning vs Deep learning

10/8/24

<u>Aspect</u>	<u>Machine learning</u>	<u>Deep learning</u>
<u>Architecture</u>	Uses the algorithm Such as decision trees SVM, KNN etc.	Uses the multi layer neural networks
<u>Data requirement</u>	Works with relatively smaller datasets.	Usually requires large datasets
<u>Feature extraction</u>	Features are usually Selected/ extracted manually	Learns features automatically
<u>Complexity</u>	Comparitively less Complex	Highly complex
<u>Computation</u>	Lower computational requirement.	Requires high Computational power

ML applications :-

1. credit card fraud detection
2. Email Spam detection.

DL applications :-

1. face recognition
2. Self driving cars.

2) Representation Learning - width and Depth.
Representation learning is a technique where a model automatically learns useful features or representations from the raw data instead of requiring humans to manually select feature.

Example :-

In image recognition, a neural network can learn
pixels $\rightarrow$ edges $\rightarrow$ shapes $\rightarrow$ objects $\rightarrow$ class.

Width :-

Width means the no. of layers in a layers.

- More neurons can learn more patterns.

- Increase model capacity.

- Too many neurons can increase computation and cause overfitting.

Depth :-

Depth means the no. of layers in a neural network.

- More layers allow to learn the data.

- Increases computational complexity.

- Very deep networks may suffer from the overfitting.

Disadvantages

Require Selection of the α parameter
performance may vary depending on α .

3. ELU

It stands for Exponential linear unit.

Formula:

$$f(x) = \begin{cases} x & x > 0 \\ \alpha(e^x - 1) & x \leq 0 \end{cases}$$

Advantages

Reduce the dying neuron problem
can improve learning in some deep networks.

Disadvantages

More computationally expensive than ReLU.

Unsupervised learning, RBM and Autoencoder.

Unsupervised learning is a type of machine learning where the model learns patterns for unlabeled data.

Examples include

- Clustering
- Feature learning
- Anomaly detection

An RBM is a neural network consisting mainly of

Visible layer $\leftrightarrow$ Hidden layer

There are no connection between neurons within the same layer.

Compare ReLU, Leaky ReLU and ELU

Activation function introduces non linearity into the neural networks, allowing them to learn complex relationships.

1. ReLU

ReLU Stands for Rectified linear unit

Formula

$$f(x) = \max(0, x)$$

Therefore

If $x > 0 \rightarrow \text{output} = x$

If $x \leq 0 \rightarrow \text{output} = 0$

Advantages:-

- Simple and Computationally efficient.
- Fast training
- widely used in deep neural networks.

Disadvantages:-

Dying ReLU problem: neurons can become inactive when they continuously receive negative inputs.

2. Leaky ReLU

Formula:-

$$f(x) = \begin{cases} x & x > 0 \\ ax & x \leq 0 \end{cases}$$

Where a is small value such as 0.01

Advantages:-

Reduces the dying ReLU problem

Allows negative values to pass through.

Autoencoder:-

An autoencoder consists of:

Input $\rightarrow$ Encoder $\rightarrow$ Hidden / Latent Representation $\rightarrow$ Decoder $\rightarrow$ output

Objectives:-

Minimize the difference b/w input and reconstructed output.

RBM VS Autoencoder.

RBM

- Probabilistic / generative model
- Learns probability distribution.
- Uses visible and hidden units
- Useful for generation

Autoencoder

- Neural-network reconstruction model
- Learns compressed representation
- Uses encoder and decoder
- Useful for compression.

Autoencoder and its uses in Deep learning

An Autoencoder is an unsupervised neural network that learns to represent input data in a compressed form and then reconstruct the original data.

Architecture

Input $\rightarrow$ Encoder $\rightarrow$ Latent

Reconstructed output

Space $\rightarrow$ Decoder $\rightarrow$

Uses of Autoencoders:-

- 1) Dimensional Reduction.
- 2) Image reconstruction
- 3) Representation learning
- 4) Denoising
- 5) Anomaly detection.
- 6) Recurrent the data.
- 7) Data Compression.
- 8) Feature extraction.

Example:-

For a noisy image

Noisy image $\rightarrow$ Encoder $\rightarrow$ Latent Features $\rightarrow$
Decoder $\rightarrow$ Clean / Reconstructed image.

An autoencoder learns to compress the image and then pass through the encoder and reconstruct the data making it useful for the feature learning, compression, denoising and anomaly detection.